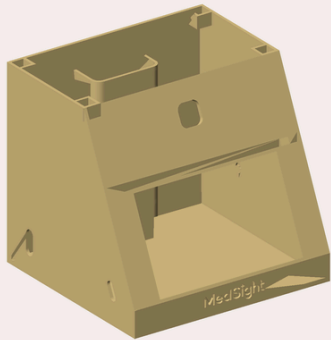
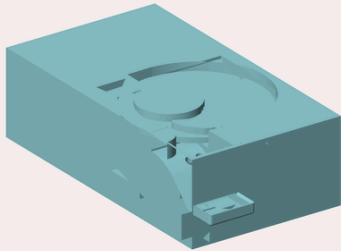
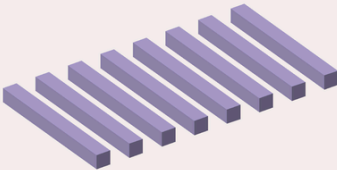


MedSight - the printed parts

01 Board enclosure164 × 153 × 152 mm	02 Pill dispenser179 × 92 × 50 mm	03 Structural pillars180 × 100 × 10 mm
		
Houses the STM32N6570-DK and presents the 5-inch screen at a readable angle. The sloped front face is why the device reads as an appliance rather than a dev board on a desk.	The mechanism. One turntable, one medication, which is where classification happens in this device: mechanically, rather than by vision.	The frame that carries the enclosure and the dispenser as one assembly.
#PARTFUNCTION	#PARTFUNCTION	#PARTFUNCTION
1Outer housing shellMain enclosure body with sloped structural walls	1Outer enclosure boxRectangular housing, open at the top over the turntable	1Structural pillar ×8Hollow rectangular-section beams
2Port / connector gridCircular openings on the side wall for USB-C and the jumper harness out to the dispenser	2Rotating pill carouselThe disc, with a helical pill track. Mounted on the 28BYJ-48 and driven half-step	
3Battery compartmentRear-mounted recess for the battery. This is what makes standalone operation a product feature rather than a demo: the battery lives inside the device rather than trailing from it	3Central drive motor hubSpindle coupling the stepper to the carousel	
4MedSight brand plateEmbossed panel on the front base face	4Dispensing chuteExit channel to the collection drawer. The chute geometry is what makes the IR counter work: pills SLIDE rather than free-fall, which is why detection pulses measure 17-46 ms instead of the few milliseconds a drop would give	
	5Output collection drawerThe external box the dose arrives in	
	6Side sensor brackets ×2Mounts holding the IR proximity module square to the chute at a repeatable height and standoff, so one pill's pulse looks like the next one's	
	7Internal divider wallVertical partition separating the mechanism from pill storage	

Every drawing and every dimension on this sheet is taken from the STL files themselves. Numbered callouts for each sub-part are in docs/HARDWARE_SPEC.md.

Process	FDM, 3D printed
Assemblies	3 (+ 8× pillar)
Largest part	164 × 153 × 152 mm
Mechanism	28BYJ-48 stepper, half-step, via ULN2003A
Pill counting	IR proximity, EXTIO, both edges
Measured pulse	17-46 ms (pills slide, not fall)
Electrical	11 jumper wires, one 5 V source, one ground
External components	none, no resistors or transistors
CAD supplied	STL to print, STEP to edit, one of each per part
Print settings	FDM, PLA, 20% infill
Still outstanding	fastener list

docs/HARDWARE_SPEC.md section 2.2 for the callout tables, section 3 for the wiring. docs/MEDSIGHT_HANDBOOK.md for everything else.

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